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Carrier Transport Equations in Semiconductors

This chapter discusses carrier transport models for electrons and holes in inorganic semiconductors.

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Introduction to Carrier Transport Models

Sentaurus Device supports several carrier transport models for semiconductors. They all can be written in the form of continuity equations, which describe charge conservation:

Equation 58

$$\nabla \cdot \vec{J}_n = q(R_{\mathrm{net},n} - G_{\mathrm{net},n}) + q\frac{\partial n}{\partial t} \qquad -\nabla \cdot \vec{J}_p = q(R_{\mathrm{net},p} - G_{\mathrm{net},p}) + q\frac{\partial p}{\partial t}$$

where:

- $R_{\mathrm{net},n}$ and $R_{\mathrm{net},p}$ are the electron and hole net recombination rate, respectively.
- $G_{\mathrm{net},n}$ and $G_{\mathrm{net},p}$ are the electron and hole net generation rate, respectively.
- $\vec{J}_n$ is the electron current density.

- $\vec{J}_p$ is the hole current density.
- n and p are the electron and hole density, respectively.

The transport models differ in the expressions used to compute $\vec{J}_n$ and $\vec{J}_p$. These expressions are the topic of this chapter. Additional equations to compute temperatures are usually also considered part of a transport model but are deferred to Chapter 9, Lattice and Carrier Temperature Equations on page 274. Models for $R_{\mathrm{net,n}}$ and $R_{\mathrm{net,p}}$ are discussed in Chapter 16, Generation–Recombination on page 525, Chapter 17, Traps and Fixed Charges on page 608, and Chapter 24, Tunneling on page 908. Boundary conditions for Equation 58 are discussed in Chapter 10, Boundary Conditions on page 289.

Depending on the device under investigation and the level of modeling accuracy required, you can select different transport models:

- **Drift-diffusion**: Isothermal simulation, suitable for low-power density devices with long active regions (see Drift-Diffusion Model).
- **Thermodynamic**: Accounts for self-heating. Suitable for devices with low thermal exchange, particularly, high-power density devices with long active regions (see Thermodynamic Model for Current Densities on page 269).
- **Hydrodynamic**: Accounts for energy transport of the carriers. Suitable for devices with small active regions (see Hydrodynamic Model for Current Densities on page 269).
- **Monte Carlo**: Solves the Boltzmann equation for a full band structure.

The numeric approach for the Monte Carlo method (and the physical models usable with it) differs significantly from the other transport models. Therefore, the Monte Carlo method is described separately (see the *Sentaurus™ Device Monte Carlo User Guide*).

For all other transport models, the solution of Equation 58 is requested by the keywords `Electron` and `Hole` in the `Solve` section. When the equation for a density is not solved in a `Solve` statement, the quasi-Fermi potential for the respective carrier type remains fixed, unless the keyword `RecomputeQFP` is present in the `Math` section and the equation for the other density is solved.

Solutions of the densities n and p are stored in the datasets `eDensity` and `hDensity`, respectively. Current densities $\vec{J}_n$ and $\vec{J}_p$ are stored in the vector datasets `eCurrentDensity` and `hCurrentDensity`, their sum is stored in `ConductionCurrentDensity`, and the total current density (including displacement current) is stored in `TotalCurrentDensity`. For an alternative representation of the total current density, see Current Potential on page 272.

Note:

Maxwell–Boltzmann statistics is the default for transport models in Sentaurus Device, and you can choose to use Fermi–Dirac statistics as described in Fermi Statistics on page 259.

Drift-Diffusion Model

The drift-diffusion model is the default carrier transport model in Sentaurus Device. For the drift-diffusion model, the current densities for electrons and holes are given by:

Equation 59

$$\vec{J}_n = \mu_n(n\nabla E_C - 1.5nkT\nabla \ln m_n) + D_n(\nabla n - n\nabla \ln \gamma_n) + \mu_n nk\left(\ln\left(\frac{n}{N_c(T)}\right) - \frac{3}{2} - \ln(\gamma_n)\right)\nabla T$$

Equation 60

$$\vec{J}_p = \mu_p(p\nabla E_V + 1.5pkT\nabla \ln m_p) - D_p(\nabla p - p\nabla \ln \gamma_p) - \mu_p pk\left(\ln\left(\frac{p}{N_v(T)}\right) - \frac{3}{2} - \ln(\gamma_p)\right)\nabla T$$

The first term takes into account the contribution due to the spatial variations of the electrostatic potential, the electron affinity, and the band gap. The remaining terms take into account the contribution due to the gradient of concentration, and the spatial variation of the effective masses m_n and m_p. For Fermi statistics, γ_n and γ_p are given by Equation 48 and Equation 49. For Boltzmann statistics, $\gamma_n = \gamma_p = 1$. Finally, the last term, the temperature gradient part, ensures that a difference in the temperature does not lead to a voltage difference or an electrical current, because, in the drift-diffusion model no thermoelectric effects are treated.

By default, the diffusivities D_n and D_p are given through the mobilities by the Einstein relation, $D_n = kT\mu_n$ and $D_p = kT\mu_p$. However, you can compute them independently (see Non-Einstein Diffusivity on page 505).

When the Einstein relation holds, the current equations can be simplified to:

Equation 61

$$\vec{J}_n = -qn\mu_n\nabla\Phi_n = \mu_n\left[(n\nabla E_c - 1.5nkT\nabla \ln m_n) + kT(\nabla n - n\nabla \ln \gamma_n) + nk\left(\ln\left(\frac{n}{N_c(T)}\right) - \frac{3}{2} - \ln(\gamma_n)\right)\nabla T\right]$$

Equation 62

$$\vec{J}_p = -qp\mu_p\nabla\Phi_p = \mu_p\left[(p\nabla E_v + 1.5pkT\nabla \ln m_p) - kT(\nabla p - p\nabla \ln \gamma_p) - pk\left(\ln\left(\frac{p}{N_v(T)}\right) - \frac{3}{2} - \ln(\gamma_p)\right)\nabla T\right]$$

where Φ_n and Φ_p are the electron and hole quasi-Fermi potentials, respectively (see Quasi-Fermi Potential With Boltzmann Statistics on page 258).

You can use the drift-diffusion model with a lattice temperature equation, but it is not mandatory. It is not possible to use the drift-diffusion model for a particular carrier type and to solve the carrier temperature for the same carrier type; the hydrodynamic model is required for that (see Hydrodynamic Model for Current Densities on page 269).

Thermodynamic Model for Current Densities

In the thermodynamic model [1], the relations Equation 61 and Equation 62 are generalized to include the temperature gradient as a driving term:

Equation 63

$$\vec{J}_n = -qn\mu_n(\nabla\Phi_n + P_n\nabla T)$$
$$= \mu_n\left[n\nabla\left(E_c - 1.5nkT\nabla\ln(m_n) + kT(\nabla n - n\nabla\ln\gamma_n) + nk\left(\ln\left(\frac{n}{N_C(T)}\right) - \frac{3}{2} - \ln(\gamma_n) - \frac{q}{k}P_n\right)\nabla T\right)\right]$$

Equation 64

$$\vec{J}_p = -qp\mu_p(\nabla\Phi_p + P_p\nabla T)$$
$$= \mu_p\left[p\nabla E_v + 1.5pkT\nabla\ln(m_p) - kT(\nabla p - p\nabla\ln\gamma_p) - pk\left(\ln\left(\frac{p}{N_V(T)}\right) - \frac{3}{2} - \ln(\gamma_p) - \frac{q}{k}P_p\right)\nabla T\right]$$

where P_n and P_p are the absolute thermoelectric powers [2] (see Thermoelectric Power on page 1146), and T is the lattice temperature.

The model differs from drift-diffusion when the lattice temperature equation is solved (see Thermodynamic Model for Lattice Temperature on page 279). However, it is possible to solve the lattice temperature equation even when using the drift-diffusion model.

The thermodynamic model is used by default. However, optionally, the `Thermodynamic` keyword can still be specified in the global `Physics` section. In addition, the drift-diffusion model can be enabled by explicitly disabling the thermodynamic model by specifying `-Thermodynamic` in the global `Physics` section.

Hydrodynamic Model for Current Densities

In the hydrodynamic model, current densities are defined as:

Equation 65

$$\vec{J}_n = \mu_n\left(n\nabla E_C + kT_n\nabla n - nkT_n\nabla\ln\gamma_n + \lambda_n f_n^{td}kn\nabla T_n - 1.5nkT_n\nabla\ln m_n\right)$$

Equation 66

$$\vec{J}_p = \mu_p\left(p\nabla E_V - kT_p\nabla p + pkT_p\nabla\ln\gamma_p - \lambda_p f_p^{td}kp\nabla T_p + 1.5pkT_p\nabla\ln m_p\right)$$

The first term takes into account the contribution due to the spatial variations of electrostatic potential, electron affinity, and the band gap. The remaining terms in Equation 65 and Equation 66 take into account the contribution due to the gradient of concentration, the carrier temperature gradients, and the spatial variation of the effective masses m_n and m_p. For Fermi statistics, γ_n and γ_p are given by Equation 48 and Equation 49, and $\lambda_n = F_{1/2}(\eta_n) / F_{-1/2}(\eta_n)$ and $\lambda_p = F_{1/2}(\eta_p) / F_{-1/2}(\eta_p)$ (with η_n and η_p from Equation 50 and Equation 51); for Boltzmann statistics, $\gamma_n = \gamma_p = \lambda_n = \lambda_p = 1$.

The thermal diffusion constants f_n^{td} and f_p^{td} are available in the `ThermalDiffusion` parameter set. They default to zero, which corresponds to the Stratton model [3][4].

Equation 65 and Equation 66 assume that the Einstein relation $D = \mu kT$ holds (D is the diffusion constant), which is true only near the equilibrium. Therefore, Sentaurus Device provides the option to replace the carrier temperatures in Equation 65 and Equation 66 by $gT_c + (1-g)T$, an average of the carrier temperature (T_n or T_p) and the lattice temperature. This can be useful in devices where the carrier diffusion is important. The coefficient g for electrons and holes can be specified in the `ThermalDiffusion` parameter set. It defaults to one.

To activate the hydrodynamic model, specify the keyword `Hydrodynamic` in the global `Physics` section. If only one carrier temperature equation is to be solved, then `Hydrodynamic` must be specified with an option, either `Hydrodynamic(eTemperature)` or `Hydrodynamic(hTemperature)`.

Note:

The calculation of λ_n and λ_p employ approximations for the Fermi integrals that are more accurate than those used in previous releases. This change can lead to different results where the `Fermi` and `Hydrodynamic` keywords are specified in the global `Physics` section. For backward compatibility, the Fermi integrals used in previous releases can be selected by specifying `-WithConsistentStatistics` in the global `Physics` section.

Numeric Parameters for Current Densities

Occasionally, if the drift-diffusion model or the thermodynamic model is solved together with a lattice temperature equation, the thermal diffusion part $nk \ln(n / N_c(T))\nabla T$ or $pk \ln(p / N_v(T))\nabla T$ in Equation 61 and Equation 62 or Equation 63 and Equation 64 can cause convergence issues for small densities. For this reason, you have the option to introduce a small reference density to stabilize the simulation as described in Interpolation of the GradQuasiFermi Driving Force on page 502. That means, you get

$nk \ln\left(\left(n+n_0\right) / N_c(T)\right)\nabla T$ or $pk \ln\left(\left(p+p_0\right) / N_v(T)\right)\nabla T$ where the value of n_0 and p_0 can be specified in the Math section:

```
RefDens_eTEP=<float>
RefDens_hTEP=<float>
```

In addition, the reference densities are applied to the corresponding terms in the analytic formulas for thermoelectric power in the Equation 1168, Equation 1169, Equation 1170, and Equation 1171.

Numeric Parameters for Continuity Equation

Carrier concentrations must never be negative. If during a Newton iteration a concentration erroneously becomes negative, then Sentaurus Device applies damping procedures to make it positive. The concentration that finally results from a Newton iteration is limited to a value that can be specified in cm^{-3} by `DensLowLimit=<float>` in the global `Math` section (default: 10^{-100} cm^{-3}). The value specified by `DensLowLimit` cannot be smaller than $10^{-250}cm^{-3}$.

If a smaller value of the density limit is required, then specify `ExtendedPrecision` (see Extended Precision) and use the `DensLowLimitPower=<int>` parameter in the global `Math` section (default: –100). The value specified by `DensLowLimitPower` is restricted based on the `ExtendedPrecision` type, with the long double (80-bit) type having a minimum of –4500, and all other floating-point types a minimum of –250.

The resulting lowest carrier concentration limit is set to the minimum between `DensLowLimit=<float>` and $10^{\texttt{DensLowLimitPower}}$ cm^{-3}.

Note:

For very-low device temperatures (below 50 K), the carrier concentration might be extremely small and require the use of `ExtendedPrecision`.

Numeric Approaches for Contact Current Computation

By default, Sentaurus Device computes the contact current as the sum of the integral of the current density over the surface of the doping well associated with the contact and the integral of the charge generation rate over the volume of the doping well. The following alternative, mutually exclusive approaches can be selected in the global `Math` section:

- `CurrentWeighting` activates a domain-integration method that uses a solution-dependent weighting function to minimize numeric errors (see[5] for details).
- `DirectCurrent` activates computation of the current as the surface integral of the current density over the contact area.

Note:

The current that flows into circuit nodes in mixed-mode simulations is always computed with the `DirectCurrent` approach, regardless of the specification in the `Math` section.

Current Potential

The total current density:

Equation 67

$$\vec{J} = \vec{J}_n + \vec{J}_p + \vec{J}_{\mathrm{D}}$$

satisfies the conservation law $\nabla \cdot \vec{J} = 0$.

Consequently, $\vec{J}$ can be written as the curl of a vector potential $\vec{W}$:

Equation 68

$$\vec{J} = \nabla \times \vec{W}$$

In a 2D simulation, $\vec{W}$ only has a nonzero component along the z-axis, which is written simply as $W_z = W$.

The total current density is then given by:

Equation 69

$$\vec{J} = \begin{bmatrix} \frac{\partial W}{\partial y} \\ -\frac{\partial W}{\partial x} \end{bmatrix}$$

The function W has the following important properties:

- The contour lines of W are the current lines of $\vec{J}$.
- The difference between the values of W at any two points equals the total current flowing across any line linking these points.

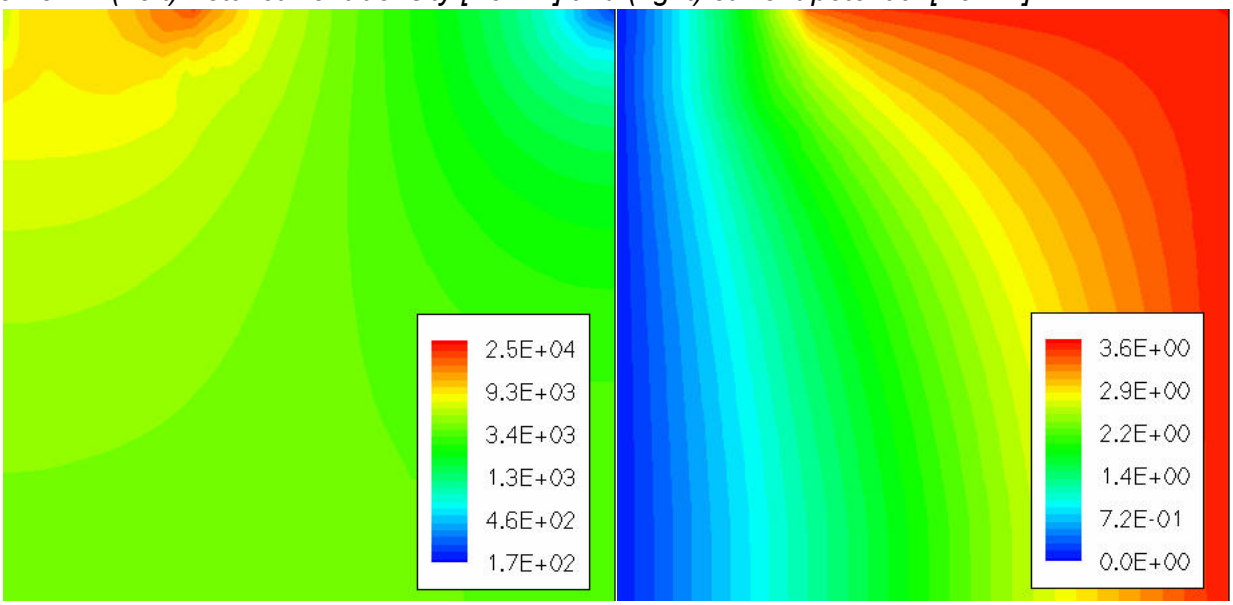
Sentaurus Device computes the 2D current potential according to the approach proposed by Palm and Van de Wiele [6].

To visualize the results, add the keyword `CurrentPotential` to the `Plot` section:

```
Plot { CurrentPotential }
```

Figure 18 displays plots of the total current density and current potential for a simple square device.

Figure 18 (Left) Total current density [Acm^{-2}] and (right) current potential [Acm^{-1}]



References

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